# EK-01 Scoring Criteria, Skill Matching and Experience Relevance: EVA-005, EVA-001, EVA-004

Document ID: EK-01 | Category C: Evaluation knowledge base

## 1. Document metadata

| Field | Value |
|---|---|
| Document ID | EK-01 |
| Filename | EK-01_Scoring_Criteria_and_Skill_Matching.pdf |
| Category | C - Evaluation knowledge base |
| Version | 1.0 |
| Creation date | 2026-10-14 |
| Data status | Factual methodology (illustrative default parameters) with hypothetical worked examples |
| Main topics | Evidence-based recommendation; Overall score; Citations; Missing data; Hallucination prevention; Skill matching; Evidence levels; Synonyms; Mandatory skills; Experience relevance; Project similarity; Recency; Double counting |
| Relevant skills and technologies | RAG pipeline; Vector search; All skills; Portfolio evaluation |
| Relevant entity IDs | EVA-005, EVA-001, EVA-004, TEC-004, EVA-002, EVA-003, EVA-006, FRE-001, TEC-001, TEC-002, PRJ-001, PRJ-002, PRJ-006 |

Purpose and scope: this PDF consolidates 3 source documents (EVA-005 to EVA-004). Each source keeps its own ID in every section heading, so chunks stay attributable, for example [DOC:EVA-005/...].

## 2. Executive summary

The overall score is a deterministic weighted average computed by code, not by the language model. The assistant explains the stored scores using retrieved evidence, cites every factual claim, reports missing data and conflicts, and never invents project history, ratings or numbers. Skill matching maps each job skill to the strongest available evidence type and gives credit by evidence strength. Direct verified evidence earns full credit, indirect or unsupported evidence earns partial credit and missing evidence earns none, but missing evidence is always reported. This PDF contains 3 source documents in total.

## 3. EVA-005: EVA-005 Evidence-Based Recommendation and Overall Scoring

**Source document ID: EVA-005** | Data status: Factual methodology (illustrative default parameters) with hypothetical worked examples

Purpose and scope: define the overall scoring framework, how missing data and hard constraints are handled, how evidence is retrieved and cited, and which behaviors are forbidden for the language model.

### EVA-005 executive summary

The overall score is a deterministic weighted average computed by code, not by the language model. The assistant explains the stored scores using retrieved evidence, cites every factual claim, reports missing data and conflicts, and never invents project history, ratings or numbers.

## EVA-005 - Default scoring framework

| Criterion | Default weight | Method document |
|---|---|---|
| Technical skill match | 30% | EVA-001 |
| Budget compatibility | 25% | EVA-002 |
| Deadline compatibility | 20% | EVA-003 |
| Relevant experience | 15% | EVA-004 |
| Reviews and reputation | 10% | QUA-004 |

All criterion scores range from 0 to 100. Overall score = sum of (criterion weight x criterion score). The framework is an illustrative baseline, not an immutable rule; recommendation quality depends on well-defined criterion scoring. When a client customizes weights, the weights are normalized: w_i = raw_i / sum(raw), and the sum must be greater than zero, otherwise the request is rejected.

## EVA-005 - Worked examples of the overall score

| Example | Criterion scores | Normalized weights used | Overall |
|---|---|---|---|
| Example 1 | skill 90; budget 80; deadline 100; experience 60; reviews 70 | skill 0.300; budget 0.250; deadline 0.200; experience 0.150; reviews 0.100 | 83.0 |
| Example 2 | skill 85; budget 100; deadline 75; experience 70; reviews n/a | skill 0.333; budget 0.278; deadline 0.222; experience 0.167 | 84.4 |
| Example 3 | skill 90; budget 80; deadline 100; experience 60; reviews 70 | skill 0.400; budget 0.300; deadline 0.100; experience 0.100; reviews 0.100 | 83.0 |
| Example 4 | skill 70; budget 90; deadline 60; experience 50; reviews 80 | skill 0.429; budget 0.286; deadline 0.143; experience 0.143 | 71.4 |

Example 1 calculation: 0.30 x 90 + 0.25 x 80 + 0.20 x 100 + 0.15 x 60 + 0.10 x 70 = 27 + 20 + 20 + 9 + 7 = 83.0. Example 2: the reviews criterion has no data, so its weight 0.10 is removed and the remaining weights (0.30, 0.25, 0.20, 0.15) are divided by 0.90; the missing criterion is not scored as 0 and the explanation states 'reputation not scored: no reviews'. Example 3: raw weights 40, 30, 10, 10, 10 sum to 100 and are normalized to 0.40, 0.30, 0.10, 0.10, 0.10. Example 4: raw weights 3, 2, 1, 1, 0 sum to 7 and the zero-weight criterion is ignored.

## EVA-005 - Mandatory requirements versus weighted preferences

Hard constraints are evaluated before ranking. Hypothetical example: criterion scores skill 95, budget 0, deadline 100, experience 90, reviews 90 give a weighted score of 71.0, yet the budget is above the hard limit so the status is INELIGIBLE_HARD_CONSTRAINT and the proposal is not recommended even though its weighted score looks high. Status values: ELIGIBLE, ELIGIBLE_WITH_FLAGS, HOLD_PENDING_EVIDENCE (a mandatory skill lacks direct evidence) and INELIGIBLE_HARD_CONSTRAINT (hard budget or hard deadline exceeded). Eligible proposals are ranked first; held proposals are listed separately with the evidence needed.

## EVA-005 - Missing data and confidence

| Situation | Treatment |
|---|---|
| A criterion has no data (for example no reviews) | Exclude the criterion, renormalize remaining weights, state it explicitly |
| Skill evidence missing for a required skill | Credit 0 for that skill and report 'no evidence found' |
| Conflicting evidence (profile and case study disagree) | Prefer the project case study; report the conflict |
| Evidence older than three years | Report the date; lower recency factor handles weight |

Confidence label: High when all five criteria are scored and no flags concern evidence quality; Medium when one criterion is missing or flags exist; Low when two or more criteria are missing. Confidence describes data completeness, not the quality of the freelancer.

## EVA-005 - Evidence retrieval procedure

- Identify entities: freelancer ID, proposal ID, job ID and skills from the database record.

- Filter by metadata (document_id, category, freelancer_id, project_id, skills) before semantic ranking.
- Run hybrid retrieval: semantic similarity plus keyword match, then merge and rerank.
- Retrieve top K chunks (for example K = 5 to 8) from profiles, project case studies, technical guidance and methodology documents.
- Drop chunks that do not mention the entity in question; never attribute a project chunk to a freelancer not listed in it.
- Pass the stored scores and retrieved chunks to the language model with an instruction to explain, not recalculate.

### EVA-005 - Evidence references and source attribution

Each cited chunk gets a reference of the form [DOC:<document_id>/<section title>], for example [DOC:FRE-003/Skill evidence table for FRE-003] or [DB:proposal PRP-001]. Database facts (price, deadline, ratings, scores) are cited as [DB:<table>/<id>]. Every material factual claim in a recommendation must carry at least one reference. Claims without a retrievable source are removed or rewritten as 'not found'.

Example of an acceptable sentence: 'FRE-002 demonstrated Angular competency in PRJ-001 by implementing the product filtering component and its unit tests [DOC:PRJ-001/Freelancers involved and individual contributions].' Example of an unacceptable sentence: 'He did that in the project mentioned above.'

### EVA-005 - Rules that prevent invented content

- The language model must not calculate, change or round scores; it quotes stored values.
- It must not create project history, ratings, reviews, prices, deadlines or availability that are not in the database or retrieved chunks.
- If no evidence is found, it states: 'No evidence for <skill> was found for <freelancer ID> in the knowledge base.'
- It separates documented facts, indirect evidence, unsupported claims and its own interpretation.
- It states uncertainty and the number of reviews and records behind each statement.
- It must not recommend by protected or irrelevant characteristics (EVA-006).

### EVA-005 - Explanation template

Proposal PRP-ID, overall score X (stored). Strengths: criteria with high scores plus cited evidence. Weaknesses: criteria with low scores plus cited reasons. Risks and flags: list with severity. Missing information: list. Confidence: label. Client decision note: recommendation supports, and does not replace, client judgement.

### EVA-005 - Retrieval evaluation

Keep a test set of questions with expected document IDs, for example 'Which freelancer has documented experience with Spring Boot and PostgreSQL?' expecting FRE-003 and FRE-001 profile documents and PRJ-001 and PRJ-002 case studies. Measure Recall@K, MRR (mean reciprocal rank) and nDCG@K, and re-run after changes to chunking, embeddings or documents.

### EVA-005 - Summary of key findings

- Overall scoring is deterministic code; the LLM explains and cites.
- Missing criteria are excluded with renormalized weights, never silently scored 0.
- Hard constraints can make a high-scoring proposal ineligible.
- Every factual claim needs a source reference, otherwise it is removed.

### EVA-005 - Keywords and alternative terminology

overall score; weighted score; weights normalization; evidence-based recommendation; citation; source attribution; hallucination; missing data; confidence; hard constraint; ranking; RAG explanation; Recall@K; MRR; nDCG

# 4. EVA-001: EVA-001 Skill Matching Methodology

**Source document ID: EVA-001** | Data status: Factual methodology (illustrative default parameters) with hypothetical worked examples

Purpose and scope: define a reproducible rule for converting a job's required and optional skills and a freelancer's evidence into a 0 to 100 technical skill match score.

## EVA-001 executive summary

Skill matching maps each job skill to the strongest available evidence type and gives credit by evidence strength. Direct verified evidence earns full credit, indirect or unsupported evidence earns partial credit and missing evidence earns none, but missing evidence is always reported.

## EVA-001 - Mapping job requirements to skills

Each job lists required skills and optional skills. A required skill may be marked mandatory. Required skills have weight 2 and optional skills weight 1 in the illustrative default. Skill names are normalized to canonical names (Angular, Spring Boot, PostgreSQL, Docker, Kubernetes, CI/CD, Python, RAG pipeline, Flutter and so on) before matching.

## EVA-001 - Synonyms and related competencies

| Canonical skill | Synonyms treated as the same skill |
|---|---|
| Angular | Angular 2+; Angular framework; Angular CLI |
| TypeScript | TS |
| Spring Boot | Spring; Spring Framework 6; Java backend |
| PostgreSQL | Postgres; relational database; PSQL |
| Docker | containers; containerization |
| Kubernetes | K8s; container orchestration |
| CI/CD | continuous integration; continuous delivery; GitHub Actions; pipelines |
| Terraform | infrastructure as code; IaC |
| RAG pipeline | retrieval-augmented generation; RAG |
| Vector search | ChromaDB; embeddings search; semantic search |
| Python | Python 3 |
| scikit-learn | sklearn; classical machine learning |

Related (not identical) skills give indirect evidence only. Examples: Docker is related to Kubernetes; Python is related to FastAPI; ChromaDB experience relates to vector search. Related-skill evidence is labelled indirect and must never be reported as direct experience.

| Skill | Verified related skills that give indirect evidence |
|---|---|
| Kubernetes | Docker |
| RxJS | Angular |
| NgRx | RxJS |
| TypeScript | Angular |
| FastAPI | Python |
| scikit-learn | PyTorch |
| PyTorch | scikit-learn |
| RAG pipeline | Vector search |
| Vector search | RAG pipeline |
| Dart | Flutter |

| Skill | Verified related skills that give indirect evidence |
|---|---|
| REST API design | Spring Boot; FastAPI |
| Spring Boot | Spring Security |
| Integration testing | Unit testing |
| Unit testing | Integration testing |
| Terraform | AWS |

## EVA-001 - Evidence types and credit

| Evidence type | Definition | Credit |
|---|---|---|
| Direct | Verified project artifact at or above the required level | 1.00 |
| Direct, lower level | Verified artifact but below the required level | 0.60 |
| Indirect | Verified related skill at L2 or above | 0.50 |
| Unsupported claim | Stated on profile with no artifact | 0.25 |
| Missing | No evidence found | 0.00 (reported as missing) |

Required levels: mandatory skills require L3; other skills require L2. Levels follow the evidence-based scale in the freelancer profiles (L1 Familiar to L4 Advanced), not years of experience.

## EVA-001 - Skill match formula

Skill score = 100 x sum(weight x credit) / sum(weight). A mandatory skill with credit below 0.60 (that is, without direct evidence) sets the status HOLD_PENDING_EVIDENCE: the proposal is not rejected, but the client is told that the mandatory skill is not evidenced and the freelancer can supply evidence.

Distinction: a skill with credit 0 reduces the skill fraction, which is a correct reflection of missing evidence. This is different from the criterion itself having no data (for example no reviews), where the criterion is excluded and the weights are renormalized (EVA-005).

## EVA-001 - Worked example (hypothetical job, using the profile of FRE-001)

Hypothetical job: required Angular (mandatory), Spring Boot (mandatory), PostgreSQL; optional Docker and Kubernetes. Evaluated freelancer: FRE-001.

| Skill | Role in job | Evidence type for FRE-001 | Credit |
|---|---|---|---|
| Angular | required/mandatory | direct | 1.00 |
| Spring Boot | required/mandatory | direct | 1.00 |
| PostgreSQL | required | direct | 1.00 |
| Docker | optional | direct | 1.00 |
| Kubernetes | optional | indirect | 0.50 |

Calculation: required weights 2 x 3 = 6 and optional weights 1 x 2 = 2, total 8. Weighted credit sum = 2 x 1.00 + 2 x 1.00 + 2 x 1.00 + 1 x 1.00 + 1 x 0.50 = 7.50. Skill score = 100 x 7.50 / 8 = 93.8. Kubernetes receives indirect credit 0.50 because FRE-001 has verified Docker evidence at L2 (the profile also claims Kubernetes, but a claim earns only 0.25 and the higher indirect credit is used). Flags: none. Status impact: no mandatory skill is unevidenced.

## EVA-001 - Examples of weak versus strong evidence

- Strong: FRE-002 implemented product filtering, pagination and unit tests in PRJ-001 (Angular, L4 verified).
- Weak: a profile line saying 'experienced Angular developer' with no project reference.
- Indirect: FRE-005 has verified Docker L4, which gives indirect evidence for Kubernetes only when Kubernetes evidence is absent; FRE-005 also has direct Kubernetes evidence from PRJ-005, so the direct evidence is used.

### EVA-001 - Limitations, risks and uncertainty

The credit values and weights are illustrative defaults and can be tuned per platform. Synonym tables need maintenance. Evidence from the knowledge base is only as complete as the project records; missing records may reflect missing documentation rather than missing skill. The explanation must say 'no evidence found' rather than 'does not have the skill'.

### EVA-001 - Summary of key findings

- Evidence type determines credit, not years of experience.
- Mandatory skills require direct evidence or the proposal is held pending evidence.
- Missing evidence is reported, not hidden and not described as incompetence.

### EVA-001 - Keywords and alternative terminology

skill matching; skill score; required skills; optional skills; synonyms; direct evidence; indirect evidence; unsupported claim; missing evidence; mandatory skill; technical match

## 5. EVA-004: EVA-004 Experience Relevance Methodology

**Source document ID: EVA-004** | Data status: Factual methodology (illustrative default parameters) with hypothetical worked examples

Purpose and scope: define how similar a freelancer's documented projects are to a requested job, so that relevant demonstrated experience, not project count or years, drives the experience score.

### EVA-004 executive summary

Each documented project receives a relevance value from technology overlap, domain similarity, the freelancer's responsibility, complexity similarity, recency and outcome. The score uses only the top three projects with diminishing weights, so counting more projects cannot inflate it and no project is counted twice.

### EVA-004 - Project relevance formula

relevance r = (0.40 x technology overlap + 0.25 x domain similarity + 0.20 x responsibility + 0.15 x complexity similarity) x recency factor x outcome factor. If the technology overlap is 0 the project is not relevant (r = 0).

| Component | Definition (illustrative defaults) |
|---|---|
| Technology overlap | Weighted share of job skills (required weight 2, optional 1) that the freelancer evidenced in this project |
| Domain similarity | 1.0 same domain; 0.5 related domain (ecommerce, marketplace, SaaS); 0.3 AI to SaaS; otherwise 0 |
| Responsibility | Lead 1.0; contributor 0.7; minor 0.4, from the case study contribution table |
| Complexity similarity | 1.0 same level; 0.6 one level apart; 0.3 two levels apart |
| Recency factor | 1.0 within 1 year; 0.95 within 2; 0.85 within 3; 0.7 older |
| Outcome factor | 1.0 successful; 0.85 partially successful (limitations documented) |

### EVA-004 - Experience score

Sort project relevance values from highest to lowest r1, r2, r3. Experience score = 100 x (0.60 x r1 + 0.25 x r2 + 0.15 x r3). Missing slots count as 0 and trigger a 'limited relevant project evidence' flag, which is an evidence statement and not a judgement of ability. Because the weights sum to 1 and each r is below 1, no single project can reach 100.

### EVA-004 - Worked example (hypothetical job, FRE-001 projects)

Hypothetical job: required Angular (mandatory), Spring Boot (mandatory), PostgreSQL; optional Docker and Unit testing; domain ecommerce; medium complexity. Evaluation date 2026-10-14.

| Project | Tech | Domain | Resp. | Complexity | Recency | Outcome | r |
|---|---|---|---|---|---|---|---|
| PRJ-001 | 1.00 | 1.00 | 0.70 | 1.00 | 0.85 | 1.00 | 0.799 |
| PRJ-002 | 0.75 | 0.50 | 0.70 | 0.60 | 0.95 | 0.85 | 0.529 |
| PRJ-006 | 0.50 | 0.00 | 0.70 | 1.00 | 1.00 | 0.85 | 0.416 |

Sorted values: r1 = 0.799, r2 = 0.529, r3 = 0.416. Experience score = 100 x (0.60 x 0.799 + 0.25 x 0.529 + 0.15 x 0.416) = 67.4. PRJ-006 is a mobile project in which FRE-001 evidenced Spring Boot and PostgreSQL but not Angular, so its technology overlap is partial and its domain is unrelated.

### EVA-004 - Avoiding double counting

- One project contributes once, even if it demonstrates several skills.
- A team project is credited to a freelancer only for that freelancer's documented contribution.
- Reviews that refer to the same project are not additional experience; they feed the reputation criterion only (QUA-004).
- Years of experience are not used as a substitute for project evidence.

### EVA-004 - Limitations and uncertainty

Project records are synthetic and sparse. The similarity weights are defaults and should be tuned with evaluation data. A freelancer with few documented projects may still be capable; the explanation must say 'limited documented evidence'.

### EVA-004 - Summary of key findings

- Experience is relevance of documented projects, not their count or years.
- Only the top three projects count with weights 0.60, 0.25 and 0.15.
- Responsibility and recency adjust relevance; partial outcomes are discounted mildly.

### EVA-004 - Keywords and alternative terminology

experience relevance; project similarity; portfolio; recency; responsibility; double counting; experience score; similar projects; relevant experience

## 6. Summary of key findings

- EVA-005: Overall scoring is deterministic code; the LLM explains and cites.
- EVA-001: Evidence type determines credit, not years of experience.
- EVA-004: Experience is relevance of documented projects, not their count or years.

## 7. Keywords and alternative terminology

overall score; weighted score; weights normalization; evidence-based recommendation; citation; source attribution; hallucination; missing data; confidence; hard constraint; ranking; RAG explanation; Recall@K; MRR; nDCG; skill matching; skill score; required skills; optional skills; synonyms; direct evidence; indirect evidence; unsupported claim; missing evidence; mandatory skill; technical match; experience relevance; project similarity; portfolio; recency; responsibility; double counting; experience score; similar projects; relevant experience